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Effects of turbulent flame development on thermoacoustic oscillations

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Abstract

Thermoacoustic oscillations of a turbulent flame stabilized behind a bluff body in a duct are considered using (1) a kinematic model of the response of the flame to oncoming flow velocity oscillations and (2) the n–τ model of the influence of a localized heat source on longitudinal acoustic waves in a duct. The kinematic model well known in the literature is extended to allow for flame speed increase with the distance from the bluff body. Such an increase is an important peculiarity of practical premixed turbulent flames developing with the distance from a flameholder. For weak, harmonic, uniform fluctuations of the oncoming flow velocity, perturbations in the flame shape and in the total heat release rate are analytically determined. Calculated results show that the development of turbulent flame speed substantially affects both the gain and the phase of the flame transfer function and, in consequence, the stability of the flame studied.
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Keywords: Thermoacoustic oscillations; Combustion instabilities; Turbulent flame development; Kinematic model

1. Introduction

Theoretical studies of combustion (thermoacoustic) instabilities are often performed by dividing this complex problem into two parts [1–3]. First, the response of a flame to specified external perturbations (e.g., flow velocity oscillations) is investigated by resolving the structure of the combustion zone using a more or less detailed combustion submodel. The goal of such a study is to model the dynamics of the total heat release rate in the flame under the influence of flow perturbations. Second, the effect of a time-dependent heat source on longitudinal acoustic waves is analyzed.

The former subproblem is often addressed by invoking a simple kinematic model of flame oscillations [2–13]. The model reduces combustion to the propagation of an infinitely thin interface (flame surface), which separates unburned gas mixture from products. The speed, S, of the self-propagation of the interface is considered to be a known quantity that does not vary in space or time during the oscillations if the equivalence ratio is constant. Accordingly, the total heat release rate is assumed to be controlled by the surface area of this interface.

The simplification of a constant flame speed does not seem to be justified for turbulent combustion, because a typical turbulent flame is a developing front; i.e., its speed grows with the time after ignition or

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Nomenclature

a	flameholder radius
b	duct radius
c	sound speed
$\mathcal{F}$	flame transfer function
k	complex wavenumber
L	duct length
L_b	distance between flameholder and duct outlet
L_u	distance between flameholder and duct inlet
n	interaction index
l_t	integral length scale of turbulence
$R_\mathrm{m} = r_\mathrm{m} + a$ and r_m	matching radii
r	radial distance
r_f	flame radius
Q	total heat release rate in flame
$\bar{Q}$	mean total heat release rate
Q'	amplitude of oscillations in total heat release rate
q	heat release rate per unit area of duct
Δq_r	heat of reaction per mass of mixture
S	flame speed
S_0	fully developed turbulent flame speed
S_1	coefficient in developing flame speed approximation
$s = S_0/U$	normalized fully developed flame speed
T	turbulent time scale
t	time
t'	flame development time
U	mean flow velocity
u	flow velocity
u'	rms turbulent velocity
v	flow velocity oscillations
x	axial distance

Greek symbols

α	coefficient in $S(t')$ approximation
β	coefficient in $S(r_\mathrm{f})$ approximation
γ	ratio of specific heats
$\eta = \rho_\mathrm{u} c_\mathrm{u}/(\rho_\mathrm{b} c_\mathrm{b})$	specific acoustic impedance ratio
ξ	axial coordinate of flame surface
$\bar{\xi}$	axial coordinate of mean flame surface
ξ'	perturbations in ξ
ρ	density
τ	time lag
Φ	modulus of flame transfer function
φ	argument of flame transfer function
Ω, Ω_m, $\Omega_{\mathrm{m},0}$, Ω_0	normalized frequencies
ω	frequency of perturbation

Subscripts

b	burned
cr	critical
f	flame
m	matching
u	unburned

with the distance from a flameholder for expanding and stationary flames, respectively [14]. To the best of the authors' knowledge, this peculiarity of premixed turbulent flames has never been addressed within the framework of the aforementioned kinematic model or in multidimensional numerical simulations of thermoacoustic instabilities.

The goal of this article is to assess the effects of the flame speed development on premixed turbulent flame oscillations. To show the importance of such effects in a clear manner, we discuss analytical results. Accordingly, to obtain analytical solutions, we consider the simplest well-known model problem, i.e., flame oscillations under the influence of uniform perturbations in the axial flow velocity in the case of a constant equivalence ratio [2,3,7,8]. Certainly, such a model is quite simple and does not address some important phenomena that can analytically be studied even within the framework of the kinematic approach, e.g., equivalence ratio oscillations [6,11] or axial velocity perturbations convected by the mean flow [9, 13] (the latter problem is relevant to thermoacoustic instabilities driven by large vortical structures shed from a flameholder [1,15]). The use of this model is justified by our primary goal of demonstrating the effects of turbulent flame speed development on thermoacoustic instabilities in a clear and simple manner. For the same reasons (simplicity and clarity), we restrict ourselves to the case of V-flames, although the kinematic approach can also be applied to conical flames [2,9,13].

Our strategy consists of using a well-known simple model and extending it solely to address turbulent flame speed development, followed by comparison of results of the original and extended models. In particular, ducted flame oscillations were recently analyzed by Dowling [3] using the kinematic model summarized in Section 2.1. We extend her approach, allowing the turbulent flame speed to increase with the distance from flameholder. A model of this process is briefly described in Section 2.2 and is discussed in detail elsewhere [14]. The extended kinematic model allows us to analytically predict the response of total heat release $Q(t)$ in the flame to weak, uniform,

harmonic perturbations in the flow velocity (see Section 2.3) and to assess the effects of the turbulent flame speed development on $Q(t)$. To assess the effects of the speed development on the stability of a combustor, the predicted $Q(t)$ is incorporated into the well-known n–τ thermoacoustic model [1,2], as discussed in Section 2.4. Although the latter thermoacoustic model is also simple, we invoke it without any modifications to focus on our primary goal. Typical results are reported in Section 3.

2. Model

2.1. Kinematic model

We consider a confined, premixed, turbulent flame stabilized in the wake of an axisymmetric center body in a cylindrical duct and study the oscillations in the total heat release rate in the flame under the influence of oncoming flow velocity perturbations (see Fig. 1).

Within the framework of the aforementioned kinematic model, the well-known G equation [16] is invoked to simulate flame dynamics. Following Dowling [3], we consider the mean flow to be one-dimensional and independent of the radial coordinate r. The influence of density variations in the flame on the flow is neglected [2,3,9,13]. In the present article, the latter simplification is invoked to follow the strategy outlined above, i.e., to introduce a single extension of Dowling's model solely to simulate turbulent flame speed development. Nevertheless, it is worth mentioning that mean flow accelerates substantially in a confined flame, and this phenomenon can also be studied analytically [17].

The aforementioned assumptions allow us to simulate flame oscillations using the simple equation

$$\frac{\partial \xi}{\partial t} = u(t) - S\left[1 + \left(\frac{\partial \xi}{\partial r}\right)^2\right]^{1/2}, \tag{1}$$

where t is time, $u(t)$ is axial flow velocity, the condition $G(x, r, t) = x - \xi(r, t) = 0$ determines the flame surface, and x is axial distance.

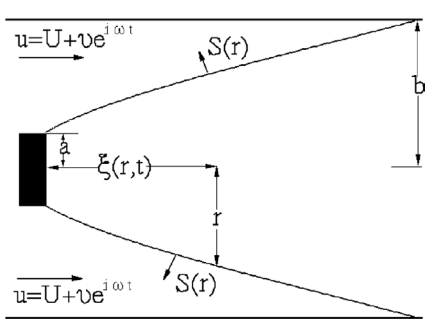

Fig. 1. Flame geometry.

In the thermoacoustic literature, the G equation and Eq. (1) are commonly applied to laminar flames, with S being the laminar flame speed [7–9,11–13]. Following Dowling [3], we apply Eq. (1) to a turbulent premixed flame, with S being a turbulent flame speed. Replacement of the laminar flame speed with the turbulent flame speed is a substantial simplification invoked (1) to reduce the real flow field, which can be decomposed into mean, oscillating but deterministic, and random (turbulent) components, to the sum of the mean and deterministic components; and, thus, (2) to make the problem analytically tractable. This simplification appears to be valid if the frequency ω of the deterministic flow velocity oscillations is low and the frequency spectrum of the turbulent velocity fluctuations is separated from ω.

The total heat release rate Q is equal to

$$Q(t) = 2\pi\rho_{\mathrm{u}}\Delta q_{\mathrm{r}} \int_a^b S\left[1 + \left(\frac{\partial \xi}{\partial r}\right)^2\right]^{1/2} r\,\mathrm{d}r, \tag{2}$$

where ρ_{u} is the density of the unburned mixture, a and b are the radii of the flameholder and duct, respectively, and Δq_{r} is the heat of reaction per unit mass of the mixture. Although oscillations in the equivalence ratio are not considered in this study, S is not constant and should be kept in the integral due to the increase in the turbulent flame speed with distance from the flameholder.

2.2. Turbulent flame speed development

To extend the kinematic model, we (1) consider flame speed to vary along the flame surface and (2) invoke the theoretical expression [14]

$$S = S_0\left\{1 + \frac{T}{t'}\left[\exp\left(-\frac{t'}{T}\right) - 1\right]\right\}^{1/2} \tag{3}$$

to model these variations. Here, $T = l_{\mathrm{t}}/u'$, l_{t}, and u' are the turbulent time, length, and velocity scales, respectively; S_0 is a fully developed turbulent flame speed; and t' designates flame development time.

The development of $S(t')$ is associated with the fact that as a flame brush grows under the influence of turbulent eddies, the flame experiences a wider range of the turbulence spectrum. Equation (3) has been obtained [14] invoking the classic theory of unsteady turbulent diffusion from a localized source in an open, constant density flow [18] to parameterize the growth of the mean thickness of a developing turbulent flame. Although (1) the pressure gradient within the flame changes turbulent scalar transport as compared with the turbulent diffusion approximation, and (2) such effects appear to be more pronounced in confined flames, the available experimental data show that the

growth of the thickness is well approximated by the Taylor theory as discussed in detail elsewhere [14] (e.g., see Figs. 21, 24, and 25 in the quoted review paper).

Equation (3) has been validated for expanding spherical flames [14]. For such flames, this submodel of the development of turbulent flame speed can straightforwardly be used, because the flame development time is simply equal to $t' = t - t_{\mathrm{i}}$ where t_{i} is the ignition time. For statistically stationary or oscillating flames, the calculation of t' is difficult. To evaluate t', one has to compute a time during which a gas volume moves from a flameholder to a point considered.

Dowling's kinematic model [3] offers the opportunity to calculate the flame development time as follows: Since the mean radial flow velocity is assumed to be equal to zero, the radial expansion of the flame is controlled solely by its speed, and the radial coordinate of the flame surface is determined from the equation

$$\frac{\mathrm{d}r_{\mathrm{f}}}{\mathrm{d}t'} = S(t'). \tag{4}$$

Equation (4) is a simplification, which holds if the radial component of the flame speed is close to S, i.e., if $S \ll u$, and the flame is long. This is generally true in a turbulent flow in a channel, where typically $S \leqslant u' \ll u$. Equations (3) and (4) supplemented with the initial condition of $r_{\mathrm{f}}(t' = 0) = a$ will be solved numerically to parameterize $t' = t'(r_{\mathrm{f}})$.

2.3. Analytical solutions for weak perturbations

Let us consider weak oncoming velocity perturbations

$$u(t) = U + v e^{i\omega t}, \tag{5}$$

with $v/U \ll 1$. Substituting $\xi(r,t) = \bar{\xi}(r) + \xi'(r)e^{i\omega t}$ into Eq. (1) and neglecting the second-order terms in v/U, one gets [3] the zero-order

$$U - S\left[1 + \left(\frac{\mathrm{d}\bar{\xi}}{\mathrm{d}r}\right)^2\right]^{1/2} = 0 \tag{6}$$

and first-order

$$\begin{aligned} i\omega\xi' &= v - S\frac{\mathrm{d}\xi'}{\mathrm{d}r}\frac{\mathrm{d}\bar{\xi}}{\mathrm{d}r}\left[1 + \left(\frac{\mathrm{d}\bar{\xi}}{\mathrm{d}r}\right)^2\right]^{-1/2} \\ &= v - \frac{\mathrm{d}\xi'}{\mathrm{d}r}\frac{S}{U}(U^2 - S^2)^{1/2} \end{aligned} \tag{7}$$

equations. Since the flame is assumed to be attached at the flameholder, the boundary conditions are as follows:

$$\bar{\xi}(a) = \xi'(a) = 0. \tag{8}$$

For arbitrary functions $v(r)$ and $S(r)$, Eqs. (7) and (8) have the solution

$$\xi' = e^{-iH(r)} \int_a^r \frac{v(y)}{S(y)(1 - S^2/U^2)^{1/2}} e^{iH(y)}\,\mathrm{d}y, \tag{9}$$

where

$$H = \omega \int_a^r \frac{\mathrm{d}y}{S(y)(1 - S^2/U^2)^{1/2}}. \tag{10}$$

The total heat release rate is decomposed into mean and oscillating components,

$$Q(t) = \bar{Q} + Q' e^{i\omega t}, \tag{11}$$

where

$$\begin{aligned} \frac{\bar{Q}}{2\pi\rho_{\mathrm{u}}\Delta q_{\mathrm{r}}} &= \int_a^b S\left[1 + \left(\frac{\mathrm{d}\bar{\xi}}{\mathrm{d}r}\right)^2\right]^{1/2} r\,\mathrm{d}r \\ &= \frac{1}{2}U(b^2 - a^2), \end{aligned} \tag{12}$$

$$\begin{aligned} \frac{Q'(t)}{2\pi\rho_{\mathrm{u}}\Delta q_{\mathrm{r}}} &= \int_a^b S\frac{\mathrm{d}\xi'}{\mathrm{d}r}\frac{\mathrm{d}\bar{\xi}}{\mathrm{d}r}\left[1 + \left(\frac{\mathrm{d}\bar{\xi}}{\mathrm{d}r}\right)^2\right]^{-1/2} r\,\mathrm{d}r \\ &= \int_a^b (v - i\omega\xi')r\,\mathrm{d}r. \end{aligned} \tag{13}$$

Integration of Eqs. (9), (10), and (13) can be performed numerically. To obtain an analytical expression for $\xi'(r)$, let us (1) consider the case of uniform velocity perturbations (v does not depend on r) and (2) approximate Eq. (3) with

$$S = \begin{cases} S_1\left(\frac{t'}{T}\right)^{\alpha} & \text{if } t' \leqslant T(S_0/S_1)^{1/\alpha}, \\ S_0 & \text{if } t' > T(S_0/S_1)^{1/\alpha}. \end{cases} \tag{14}$$

In the particular case of $S_1 = S_0/\sqrt{2}$ and $\alpha = 1/2$, the upper line corresponds to the asymptotic behavior of Eq. (3) at $t' \to 0$. In this case, however, Eq. (14) markedly overestimates flame speed development at $t'/T = O(1)$. The use of $S_1 \neq S_0/\sqrt{2}$ and $\alpha \neq 1/2$ in Eq. (14) allows us to consider a more general case and to approximate Eq. (3) better at $t'/T = O(1)$.

By substituting Eq. (14) into Eq. (4) and integrating, we obtain

$$S = \begin{cases} S_0\left[\frac{r_{\mathrm{f}} - a}{r_{\mathrm{m}}}\right]^{\beta} & \text{if } (r_{\mathrm{f}} - a) \leqslant r_{\mathrm{m}}, \\ S_0 & \text{if } (r_{\mathrm{f}} - a) > r_{\mathrm{m}}, \end{cases} \tag{15}$$

where $\beta = \alpha/(1 + \alpha)$ and

$$r_{\mathrm{m}} = \frac{TS_1}{1 + \alpha}\left(\frac{S_0}{S_1}\right)^{1/\beta}. \tag{16}$$

If the flame speed is approximated with Eq. (15) and $\beta = 1/3$ ($\alpha = 1/2$), the solutions to Eqs. (6) and (7) are

$$\bar{\xi} = \begin{cases} r_{\mathrm{m}}/s^3 - [(r_{\mathrm{m}}/s^3)^{2/3} - (r-a)^{2/3}]^{3/2} \\ \quad \text{if } (r_{\mathrm{f}} - a) \leqslant r_{\mathrm{m}}, \\ \bar{\xi}_{\mathrm{m}} + (r - R_{\mathrm{m}})(1-s^2)^{1/2}/s \\ \quad \text{if } (r_{\mathrm{f}} - a) > r_{\mathrm{m}}, \end{cases} \tag{17}$$

and

$$1 - \frac{i\omega\xi'}{v} = \begin{cases} \exp\Bigg[-i\frac{3\omega r_{\mathrm{m}}}{Us^3}\Bigg(1 - \Big[1 \\ \quad - s^2\big(\frac{r-a}{r_{\mathrm{m}}}\big)^{2/3}\Big]^{1/2}\Bigg)\Bigg] \\ \quad \text{if } (r_{\mathrm{f}} - a) \leqslant r_{\mathrm{m}}, \\ \big[1 - \frac{i\omega\xi'_{\mathrm{m}}}{v}\big]\exp\big[-i\frac{\omega(r-R_{\mathrm{m}})}{Us(1-s^2)^{1/2}}\big] \\ \quad \text{if } (r_{\mathrm{f}} - a) > r_{\mathrm{m}}, \end{cases} \tag{18}$$

respectively, where $s \equiv S_0/U$ is the normalized fully developed flame speed, $R_{\mathrm{m}} = a + r_{\mathrm{m}}$, $\bar{\xi}_{\mathrm{m}} = \bar{\xi}(R_{\mathrm{m}})$, and $\xi'_{\mathrm{m}} = \xi'(R_{\mathrm{m}})$.

The amplitude Q' of the heat release rate oscillations is equal to

$$\begin{aligned} \frac{Q'(t)}{2\pi\rho_{\mathrm{u}}\Delta q_{\mathrm{r}}} = v\int_a^{R_{\mathrm{m}}} \exp\Bigg\{-i\frac{3\omega r_{\mathrm{m}}}{Us^3}\Bigg(1 - \Bigg[1 \\ - s^2\Big(\frac{r-a}{r_{\mathrm{m}}}\Big)^{2/3}\Bigg]^{1/2}\Bigg)\Bigg\} r\,\mathrm{d}r \\ + v\Big(1 - \frac{i\omega\xi'_{\mathrm{m}}}{v}\Big)\frac{b - R_{\mathrm{m}}}{i\Omega_{\mathrm{m}}} \\ \times\Bigg\{\frac{b - R_{\mathrm{m}}}{i\Omega_{\mathrm{m}}}[1 - e^{-i\Omega_{\mathrm{m}}}] \\ - be^{-i\Omega_{\mathrm{m}}} + R_{\mathrm{m}}\Bigg\}, \end{aligned} \tag{19}$$

where

$$\Omega_{\mathrm{m}} = \frac{\omega(b - R_{\mathrm{m}})}{Us(1-s^2)^{1/2}}. \tag{20}$$

Obviously, if $R_{\mathrm{m}} > b$, the first term on the right hand side (RHS) of Eq. (19) should be integrated from a to b and the second term should be skipped.

If $\beta \neq 1/3$, the expressions

$$\bar{\xi} = \begin{cases} \frac{r_{\mathrm{m}}}{s(1-\beta)}\big(\frac{r-a}{r_{\mathrm{m}}}\big)^{1-\beta} & \text{if } (r_{\mathrm{f}} - a) \leqslant r_{\mathrm{m}}, \\ \frac{r_{\mathrm{m}}}{s(1-\beta)} + \frac{r - r_{\mathrm{m}}}{s} & \text{if } (r_{\mathrm{f}} - a) > r_{\mathrm{m}}, \end{cases} \tag{21}$$

and

$$1 - \frac{i\omega\xi'}{v} = \begin{cases} \exp\big[-i\frac{\omega r_{\mathrm{m}}}{(1-\beta)S_0}\big(\frac{r-a}{r_{\mathrm{m}}}\big)^{1-\beta}\big] \\ \quad \text{if } (r_{\mathrm{f}} - a) \leqslant r_{\mathrm{m}}, \\ \exp\big[-i\frac{\omega r_{\mathrm{m}}}{(1-\beta)sU}\big]\exp\big[-i\frac{\omega(r-R_{\mathrm{m}})}{Us}\big] \\ \quad \text{if } (r_{\mathrm{f}} - a) > r_{\mathrm{m}}, \end{cases} \tag{22}$$

satisfy

$$U = \frac{\mathrm{d}\bar{\xi}}{\mathrm{d}r}, \qquad i\omega\xi' = v - S\frac{\mathrm{d}\xi'}{\mathrm{d}r}, \tag{23}$$

which result from Eqs. (6) and (7) as $S/U \to 0$. Thus, if $\beta \neq 1/3$ and $S \ll U$, the amplitude of heat release rate oscillations is equal to

$$\begin{aligned} \frac{Q'(t)}{2\pi\rho_{\mathrm{u}}\Delta q_{\mathrm{r}}} = v\int_a^{R_{\mathrm{m}}} \exp\Bigg[-i\frac{\omega r_{\mathrm{m}}}{(1-\beta)S_0} \\ \times\Big(\frac{r-a}{r_{\mathrm{m}}}\Big)^{1-\beta}\Bigg] r\,\mathrm{d}r \\ + v\Big(1 - \frac{i\omega\xi'_{\mathrm{m}}}{v}\Big)\frac{b - R_{\mathrm{m}}}{i\Omega_{\mathrm{m},0}} \\ \times\Bigg\{\frac{b - R_{\mathrm{m}}}{i\Omega_{\mathrm{m},0}}[1 - e^{-i\Omega_{\mathrm{m},0}}] \\ - be^{-i\Omega_{\mathrm{m},0}} + R_{\mathrm{m}}\Bigg\}, \end{aligned} \tag{24}$$

where

$$\Omega_{\mathrm{m},0} = \frac{\omega(b - R_{\mathrm{m}})}{Us}, \tag{25}$$

and Eq. (12) holds. Again, if $R_{\mathrm{m}} > b$, the first term on the RHS of Eq. (24) should be integrated from a to b and the second term should be skipped.

The above analytical results can be rewritten in the form

$$\frac{Q'}{\bar{Q}} = \frac{v}{U}\mathcal{F} = \frac{v}{U}\Phi e^{i\varphi}, \tag{26}$$

where Φ and φ are the modulus (gain) and argument (phase), respectively, of the complex flame transfer function $\mathcal{F}$.

In the case of $\beta = 1/3$ (see Eqs. (12) and (19)), the transfer function is equal to

$$\begin{aligned} \mathcal{F} = \frac{2}{b^2 - a^2}\int_a^{R_{\mathrm{m}}} \exp\Bigg\{-i\frac{3\omega r_{\mathrm{m}}}{Us^3}\Bigg(1 - \Bigg[1 \\ - s^2\Big(\frac{r-a}{r_{\mathrm{m}}}\Big)^{2/3}\Bigg]^{1/2}\Bigg)\Bigg\} r\,\mathrm{d}r \\ + \Big(1 - \frac{i\omega\xi'_{\mathrm{m}}}{v}\Big)\frac{2(b - R_{\mathrm{m}})}{i\Omega_{\mathrm{m}}(b^2 - a^2)} \\ \times\Bigg\{\frac{b - R_{\mathrm{m}}}{i\Omega_{\mathrm{m}}}[1 - e^{-i\Omega_{\mathrm{m}}}] \\ - be^{-i\Omega_{\mathrm{m}}} + R_{\mathrm{m}}\Bigg\}, \end{aligned} \tag{27}$$

where the first integral, I_1, is numerically calculated. As $T \to 0$, $S \to S_0$, $R_{\mathrm{m}} \to a$, the normalized frequency Ω_{m} tends to

$$\Omega_0 = \frac{\omega(b - a)}{Us(1-s^2)^{1/2}}. \tag{28}$$

Subsequently, $\xi'_{\mathrm{m}} \to 0$, $I_1 \to 0$, and Eq. (27) reduces to the expression reported in many papers (e.g., see Eq. (2.18) in Ref. [3] or Eq. (30) with $\omega_* = -\Omega$ in Ref. [9]).

In the case of $\beta \neq 1/3$ and $S/U \to 0$, Eqs. (12), (24), and (26) yield

$$\mathcal{F} = \frac{2}{b^2 - a^2} \int_a^{R_{\mathrm{m}}} \exp\left[-i \frac{\omega r_{\mathrm{m}}}{(1-\beta) S_0} \times \left(\frac{r-a}{r_{\mathrm{m}}} \right)^{1-\beta} \right] r \, \mathrm{d}r + \left(1 - \frac{i\omega \xi'_{\mathrm{m}}}{v} \right) \frac{2(b - R_{\mathrm{m}})}{i\Omega_{\mathrm{m},0}(b^2 - a^2)} \times \left\{ \frac{b - R_{\mathrm{m}}}{i\Omega_{\mathrm{m},0}} [1 - e^{-i\Omega_{\mathrm{m},0}}] - b e^{-i\Omega_{\mathrm{m},0}} + R_{\mathrm{m}} \right\}, \tag{29}$$

where the integral, I_2, is numerically calculated. As $T \to 0$, $S \to S_0$, $R_{\mathrm{m}} \to a$, $I_2 \to 0$, and Eq. (29) reduces to the aforementioned well-known expression [3,9] rewritten in the limit of long flames ($S/U \to 0$).

2.4. Thermoacoustic submodel

To assess the effect of turbulent flame speed development on thermoacoustic oscillations, let us consider the following simple case often discussed in the literature [1,2]. Flame oscillates in a long duct, open at the exit (pressure node) and closed at the other end (velocity node), under the influence of one-dimensional plane harmonic acoustic waves that propagate along the longitudinal duct axis. For simplicity, the dissipation of the acoustic waves is neglected and the Mach number is assumed to be small. If the acoustic wavelength is much larger than the flame length, the flame can be reduced to an infinitely thin heat source. Then, the dispersion relation

$$\eta \cos(k_{\mathrm{u}} L_{\mathrm{u}}) \cos(k_{\mathrm{b}} L_{\mathrm{b}}) - \sin(k_{\mathrm{u}} L_{\mathrm{u}}) \sin(k_{\mathrm{b}} L_{\mathrm{b}})(1 + n e^{-i\omega\tau}) = 0 \tag{30}$$

can be obtained [1,2] provided that (the so-called n–τ model)

$$(\gamma - 1) \frac{q'}{\rho_{\mathrm{u}} c_{\mathrm{u}}^2} = v n e^{-i\omega\tau}. \tag{31}$$

Here, L_{u} and $L_{\mathrm{b}} = L - L_{\mathrm{u}}$ are distances from the mean flame position to the inlet and outlet of the duct, respectively, L is the duct length, $k_{\mathrm{u}} = \omega/c_{\mathrm{u}}$ and $k_{\mathrm{b}} = \omega/c_{\mathrm{b}}$ are the wavenumbers in the unburned and burned mixture, respectively, c_{u} and c_{b} are the sound speeds in the unburned and burned mixture, respectively, $\eta = \rho_{\mathrm{u}} c_{\mathrm{u}} / (\rho_{\mathrm{b}} c_{\mathrm{b}})$ is the specific acoustic impedance ratio, $q(t) = q' \exp(i\omega t)$ are the harmonic oscillations in the heat release per unit cross section area of the duct, $v \exp(i\omega t)$ are the harmonic oscillations in the flow velocity just ahead the flame, and γ is the ratio of specific heats.

If we define a time lag $\tau \equiv -\varphi/\omega$, then Eq. (26) derived above reduces to Eq. (31) with the interaction index, n, being equal to

$$n = (\gamma - 1) \frac{\Delta q_{\mathrm{r}}}{c_{\mathrm{u}}^2} \frac{b^2 - a^2}{b^2} \Phi. \tag{32}$$

Consequently, the frequency of oscillations can be determined from

$$\eta \cos(k_{\mathrm{u}} L_{\mathrm{u}}) \cos(k_{\mathrm{b}} L_{\mathrm{b}}) - \sin(k_{\mathrm{u}} L_{\mathrm{u}}) \sin(k_{\mathrm{b}} L_{\mathrm{b}}) \left[1 + (\gamma - 1) \times \frac{\Delta q_{\mathrm{r}}}{c_{\mathrm{u}}^2} \frac{b^2 - a^2}{b^2} \Phi e^{i\varphi} \right] = 0. \tag{33}$$

The combustor is unstable if the imaginary part of the complex frequency, ω, is negative.

3. Results and discussion

Different approximations for developing turbulent flame speed, used by us, are shown in lines in Fig. 2 and the corresponding mean flame shapes are drawn in Fig. 3. Symbols have been computed by numerically solving Eqs. (3) and (4) with $u' = 0.05U$, $l_{\mathrm{t}} = 0.2b$, $S_0 = 0.09U$, and $b = 2a$ (Dowling [3] reported the same values of S_0 and b/a). Curve 1 corresponds to the assumption of a constant flame

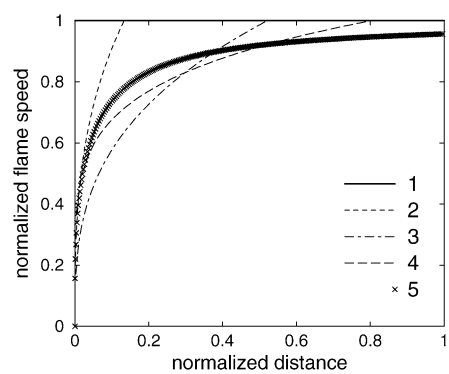

Fig. 2. Normalized flame speed, S/S_0, versus the normalized radial coordinate, $(r - a)/a$, of the flame: (1) constant flame speed ($S/S_0 = 1$); (2–4) Eqs. (15) and (16) with $S_1 = S_0/\sqrt{2}$ and $\beta = 1/3$ (2), $S_1 = 0.51 S_0$ and $\beta = 1/3$ (3), or $S_1 = 0.58 S_0$ and $\beta = 0.25$ (4); (5) numerical solution to Eqs. (3) and (4).

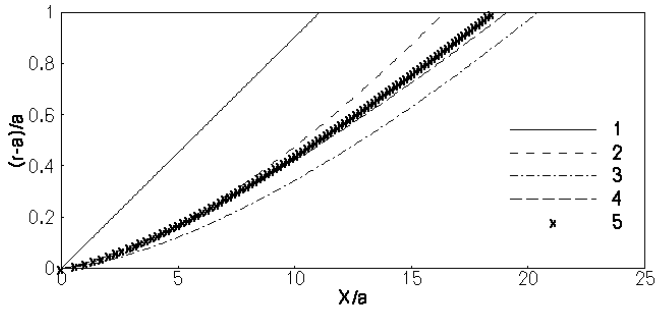

Fig. 3. Steady flame shapes obtained using the approximations of the turbulent flame speed, shown in Fig. 2.

speed ($S/S_0 = 1$). Curve 2 has been calculated using Eqs. (15) and (16) with $S_1 = S_0/\sqrt{2}$ and $\beta = 1/3$; i.e., this curve is associated with the Taylor expansion of Eq. (14). Curves 3 and 4 have been obtained using Eqs. (15) and (16) with $S_1 = 0.51S_0$, $\beta = 1/3$ and $S_1 = 0.58S_0$, $\beta = 0.25$, respectively. Curve 4 provides the best approximation of the numerical results (crosses).

The gain, Φ, and the phase, φ (in degrees), of the complex transfer functions calculated using Eqs. (26)–(29), are shown in Fig. 4. Curves 1–4 have been calculated using the approximations of $S(r)$, shown in lines 1–4, respectively, in Fig. 2.

Fig. 4a indicates that flame speed development reduces the gain at low and moderate frequencies ($\Omega_0 < 5$), but the opposite effect can be observed at higher frequencies (e.g., $\Omega_0 = 6$). Interpretation of these results is simplest in the case of $S/U \to 0$ (see curve 4) and $R_m > b$. In this case, Eq. (24) can be rewritten as

$$\frac{Q'(t)}{2\pi\rho_u\Delta q_r} = v\int_a^b \exp\left[-i\Omega_r\frac{r-a}{b-a}\right] r\,dr, \tag{34}$$

where

$$\Omega_r = \frac{\Omega_{0,0}}{1-\beta}\left(\frac{r_m}{r-a}\right)^{\beta} > \Omega_{0,0} \equiv \frac{\omega(b-a)}{S_0} = \lim_{S/U\to 0}\Omega_0. \tag{35}$$

For comparison,

$$\frac{Q'(t)}{2\pi\rho_u\Delta q_r} = v\int_a^b \exp\left[-i\Omega_{0,0}\frac{r-a}{b-a}\right] r\,dr, \tag{36}$$

in the case of a constant flame speed and $S/U \to 0$ [3]. Thus, the effects of flame speed development on $Q'(t)$ can be modeled replacing $\Omega_{0,0}$ with a higher frequency, $\Omega_r(r)$, in Eq. (36). The latter frequency is higher than $\Omega_{0,0}$ because a developing flame speed is lower than a fully developed one and the frequencies are inversely proportional to the flame speeds. After integration, the substitution of Ω_r for $\Omega_{0,0}$ results in the replacement of the original gain, $\Phi_1(\Omega_0)$, calculated using Eq. (36) (see curve 1), with the gain Φ_1 associated with a higher frequency.

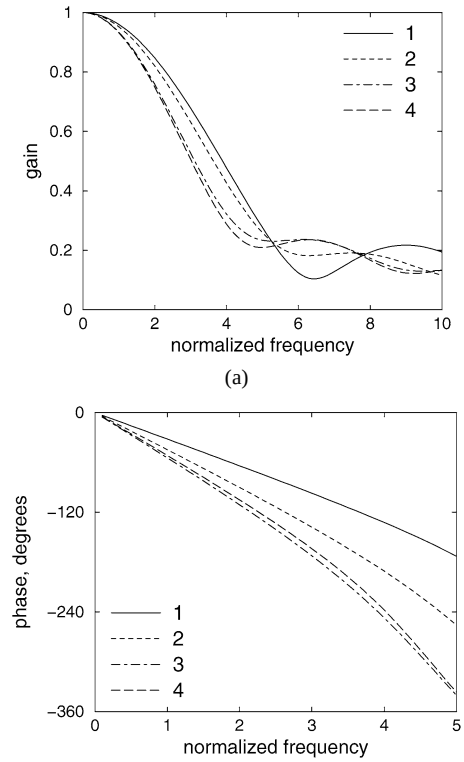

Fig. 4. Gain, Φ (a), and phase, φ (b), of the complex transfer function $\mathcal{F}$ versus the normalized frequency, Ω_0, determined by Eq. (28). Symbols are explained in Fig. 2.

At $\Omega_0 < 5$, the gain Φ_1 decreases with increasing Ω_0 and the effects discussed reduce the gain $\Phi_4 \approx \Phi_1(\Omega > \Omega_0)$ as compared with $\Phi_1(\Omega_0)$ (cf. curves 4 and 1). At higher Ω_0, the curve $\Phi_1(\Omega_0)$ exhibits oscillations (these oscillations are also shown in Fig. 5 in Ref. [9]). As a result, $\Phi_4 > \Phi_1$ in certain ranges of $\Omega_0 > 5$, the oscillations in $\Phi_4(\Omega_0)$ being

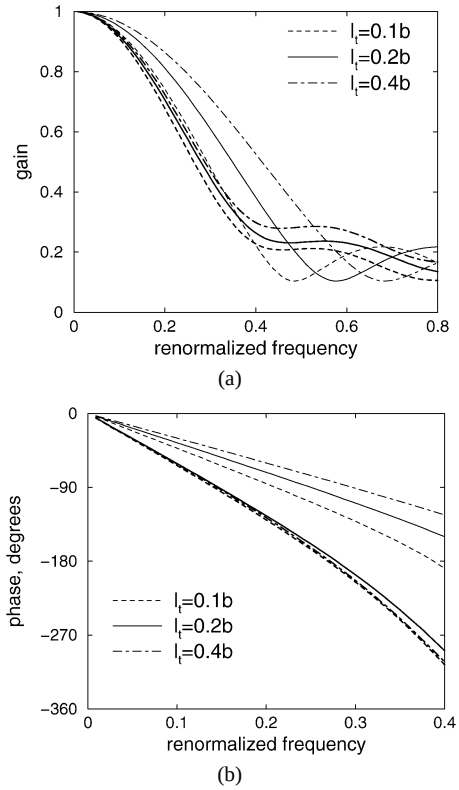

Fig. 5. Gain (a) and phase (b) of the flame transfer function versus renormalized frequency, $\omega' = a\omega/U$, at different turbulent length scales, l_t, specified in legends. All other input parameters correspond to Fig. 4. Fine curves have been calculated using a constant flame speed. Boldface curves have been obtained using Eqs. (16) and (24) with $S_1 = 0.58S_0$ and $\beta = 0.25$ (or $\alpha = 0.33$).

less pronounced due to the smoothing effects of integration.

As discussed in Section 2.1, the kinematic approach used by us (i.e., Eq. (1) with S being a turbulent flame speed) appears to be valid if the turbulence frequency spectrum is separated from the frequency of the oscillations, i.e., $\omega T \ll 2\pi$. Under the conditions of the calculations, this inequality holds if $\Omega_0 \ll 17$. Consequently, the focus of the following discussion is placed on the range of low and moderate frequencies.

In this range, flame speed development reduces the gain and, hence, the sensitivity of the flame to upstream flow perturbations. However, the development also increases the phase difference between these perturbations and heat release rate oscillations (see Fig. 4b). Under certain conditions, the latter effect can be of more importance than the gain reduction, as demonstrated in the end of this section.

A decrease in the turbulent time scale accelerates the flame development and reduces the effects discussed above as far as the functions $\Phi(\Omega_0)$ and $\varphi(\Omega_0)$ are concerned. However, the decrease in T also affects the fully developed turbulent flame speed which is increased by both u' and l_t [14]. If the time scale is decreased by increasing u' with l_t being kept constant, then S_0 is increased and Ω_0 is decreased. Both the decrease in T and the decrease in Ω_0 result in increasing gain and decreasing phase difference as far as the functions $\Phi(\omega)$ and $\varphi(\omega)$ are concerned. However, if the length scale is decreased with u' being kept constant, then the two effects (a decrease in T and an increase in Ω_0) counteract one another and the transfer function $\mathcal{F}(\omega')$ can be weakly sensitive to the length scale, as shown by the boldface lines in Fig. 5. Here, $\omega' = a\omega/U$ is a renormalized frequency.

The fine lines in Fig. 5 show the effect of the length scale on the transfer function as predicted by the original kinematic model (constant flame speed). This effect is well pronounced and controlled solely by the reduction in Ω_0 by l_t. The expression $S_0 \propto u'(T/\tau_c)^{1/4}$, which well approximates numerous experimental data [14], has been invoked to calculate the effect of l_t on $\Omega_0(S_0)$. Here, τ_c is a chemical time scale, which has been kept constant in the calculations. Boldface lines show the effect of l_t on $\mathcal{F}(\omega')$ as predicted by the kinematic model with the developing flame speed approximated with Eq. (15) (see curve 4 in Fig. 2). The effect of l_t on $\mathcal{F}$ is weakly pronounced due to the counteraction of the effects of $T = l_t/u'$ and Ω_0 on $\mathcal{F}$, as discussed above.

It is worth emphasizing that the influence of turbulence on flame oscillations is not restricted to the flame speed increase with the distance from flameholder and the dependence of the flame speed on turbulence characteristics such as u' and l_t. In addition, turbulence strongly increases the mean thickness of a premixed flame as compared with the laminar case and this phenomenon can markedly affect thermoacoustic oscillations, as discussed elsewhere [19].

Moreover, ducted turbulent flames can substantially be affected by vortex shedding from flameholder [15]. Recently, the kinematic approach has been extended to model this phenomenon by analyzing the response of a premixed flame to axial convective waves [9,13]. For this purpose, Eqs. (7) and (8) have been solved analytically in the case of a constant flame speed S but variable velocity $v = v_0 \exp(-ik\xi)$ either with the wavenumber $k = \omega/c_u$ [9] or with an arbitrary k [13]. The results reported by Schuller et al. [9] and by Preetham and Lieuwen [13] show that the flame transfer function for such perturbations differs substantially from the flame transfer function reported early for uniform perturbations [3]. In particular, the gain of the flame transfer function for the former perturbations can be larger than unity in a certain range of frequencies; i.e., a flame stabilized be-

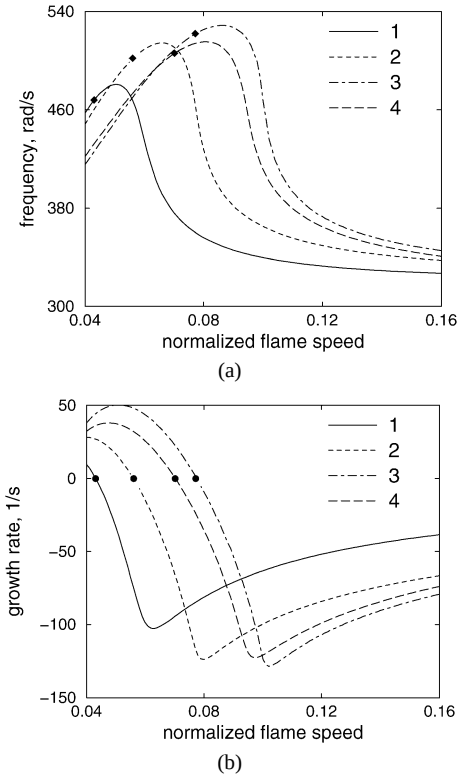

Fig. 6. Perturbation frequency (a) and growth rate (b) versus the normalized fully developed flame speed, S_0/U. Symbols are explained in Fig. 2.

hind a bluff body behaves like an amplifier of certain axial convective waves.

If one (1) considers the flame transfer function obtained by Dowling [3] in the case of a uniform perturbation and a constant flame speed to be a reference function and (2) compares the two effects (axial convective waves and flame speed increase) investigated separately in Refs. [9,13] and the present article, respectively; then, the former effect is more pronounced (cf. Figs. 5 and 6 from Ref. [9] or Fig. 11 from Ref. [13] with Fig. 4a drawn in the present paper) as far as the gain is concerned. However, such a comparison does not mean that the latter effect is of minor importance. The two phenomena can interact; e.g., the flame speed increase with the distance from flameholder can substantially affect the response of a ducted flame to axial convective waves.

To analyze this problem, Eqs. (9) and (10) with $v = v_0 \exp(-ik\xi)$ and $S = S(r)$ should be solved. Obtaining analytical expressions appears to be difficult in this case, but the integrals in Eqs. (9) and (10) can be calculated numerically, followed by numerical integration of Eq. (13). Such computations are in progress, and preliminary results show the importance of the flame speed increase in this case also.

The focus of the above discussion has been placed on the flame transfer function. To show the effects of turbulent flame speed development on the stability of a combustor in the simplest model case, we have calculated the complex frequency ω by solving Eq. (33) supplemented with different transfer functions shown in Fig. 4. The real and imaginary parts of the complex frequency, obtained for various normalized flame speeds $s = S_0/U$, are shown in Fig. 6. The ordinate axis in Fig. 6b corresponds to $-\mathrm{Im}(\omega)$. Curves 1–4 have been calculated for the first resonant frequency using the approximations of $S(r)$, shown in lines 1–4, respectively, in Fig. 2. Consequently, curves 1–4 have been drawn using the flame transfer functions shown in lines 1–4, respectively, in Fig. 4. In these computations, we have used the following values of the input parameters of the model: $u' = 0.05U$, $U = 37$ m/s, $l_t = 0.2b$, $b = 2a = 0.035$ m, $L_u = 1.18$ m, $L_b = 0.74$ m, and $\gamma = 1.4$. Dowling used the same values of U, a, b, L_u, and L_b to simulate Langhorne's experiments performed with lean (the equivalence ratio $\phi = 0.7$ and $\rho_u/\rho_b = 5.44$) ethylene–air mixtures [20]. The value of $\Delta q_r = 1.6$ MJ/kg, used by us, corresponds to the burning of this mixture with 80% efficiency (this efficiency has been reported by Dowling [3]).

Fig. 6b shows that the growth rate changes the sign from negative to positive when flame speed decreases. The value of $\mathrm{Im}(\omega) = 0$ corresponds to the transition from stable to unstable burning. Accordingly, the values of s_{cr} such that $\mathrm{Im}(\omega) = 0$ are associated with the critical condition for the occurrence of the instability. If $s < s_{cr}$, the system is unstable.

Fig. 6b indicates that the critical condition (see circles) is substantially affected by turbulent flame speed development. The frequency of the oscillations under the critical condition is also affected by this process (see diamonds in Fig. 6a). Note that these frequencies are associated with $\Omega_0 < 5$ in all cases; i.e., the flame speed development can promote the instability if $0.5 < s < 0.7$ despite the reduction in the gain Φ due to the development (see Fig. 4a).

4. Conclusions

The kinematic model of ducted flame oscillations has been extended to allow for flame speed increase with the distance from a flameholder due to the development of a premixed turbulent flame.

For weak, harmonic, uniform fluctuations of the incoming flow velocity, perturbations in the flame shape and in the total heat release rate have been determined analytically.

Calculated results show that the flame speed increase markedly affects the flame transfer function and thermoacoustic instability.

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